

Graphplay Research Program

A Companion Catalog of Dowsing-Rod Theorems, Integrations, and Higher Towers

Companion to *Graphplay: A Combinatorial Assembly Language for Quantum Primitives* (v3)

Graphplayers Crew — Graphplay v3 companion

Foreword: A Research Program in an Assembly-Language Key

The main paper presents Graphplay as a typed compiler from quantum-walk primitives (PST, mixing, search, fractional revival) to host Hamiltonians, built around a single structural primitive: the equitable partition of a Hermitian weighted graph and its small finite quotient matrix as the universal compiled object. The Lean 4 development of the five-tower spine — Set, SimpleGraph, WeightedGraph, operator algebra / coherent algebra, graphon, categorical filtered colimit — supports the three load-bearing theorems at each tower: spectral subset, cell-uniform restriction, and finite-dimensional quotient reduction.

What the main paper’s research-program section (§9) compresses into a one-page list, this companion expands into a working catalog of **open theorems**. Eighteen further Lean modules (ten “dowsing-rod” files under Graphplay/Dowsing/, eight “integration” files under Graphplay/Integrations/) plus the two higher towers Tower6.lean and Tower7.lean constitute a research program: each file fixes a precise statement that the spine **would prove** if certain identified holes (coherent-algebra commutativity, sheaf-of- $\ast$ -algebras infrastructure, ∞ -categorical Mathlib library) were closed.

Status note (this revision). An earlier draft of this companion described the whole catalog with a single blanket sentence — “statements complete, proofs sorried.” That is now stale, and we replace it with an itemized status per file. A substantial fraction of the headline content is now *proven axiom-clean* (e.g. D4’s equitable-coherent dictionary, D5’s quantum-homomorphism lift, I7’s chiral $\equiv U(1)$ gauge equivalence with zero sorry, the new TransformerDSL compiler-correctness theorem). A further band is *de-stubbed but honestly sorried at one deep step* (Tower-4 graphon spectral timing, MERA exactness, RMT thermalizing-yet-PST). The genuinely *open mathematical frontier* is the two halves of Conjecture 9.3 and the higher towers (6/7), which remain honest scaffolds awaiting ∞ -categorical / sheaf-of- $\ast$ -algebras Mathlib. We flag each item’s actual state below. Throughout, the statements typecheck against the rest of Graphplay; what has changed is how many of the proofs are now real.

The metaphor we use throughout is the **dowsing rod**. A dowsing-rod file picks out a thesis-sized open problem, formulates it as a Lean theorem (or `def : Prop`) against the existing spine, and includes worked examples sufficient to convince the reader that the statement is the right one. It is the dual of a “proof skeleton”: rather than asserting “we have proved this,” each file asserts “this is what would have to be proved, and here are the surfaces along which the spine would carry the proof.” The falsifiable Conjecture 9.3 (Bose–Mesner $\equiv$ chiral $\equiv$ graphon) is the flagship dowsing-rod result; the other nine are siblings, each at a comparable level of definiteness.

The integration files extend the metaphor outward: each is a research-program seed pointing to a neighboring field (TQFT, RMT, tensor networks, optimal transport, Hodge theory, lattice gauge theory, mean-field control, Weisfeiler–Leman refinement) where the spine applies and where, we contend, the equitable-partition idiom is currently underexploited. They are not proofs; they are **integration points**, identifying the precise places at which Graphplay’s primitives meet — and, we believe, generate — nontrivial content in the receiving field.

We present each item below in 1–2 paragraphs. The intended reader is familiar with the spine of the main paper and wants a fast survey of the catalog — enough to decide where to dig next.

1. Dowsing-Rod Theorems (D1--D10)

*The ten files under Graphplay/Dowsing/ each formulate a single thesis-sized open theorem at the boundary of the spine. Each item below carries a **Status** line recording its current Lean state: which headline statements are now proven axiom-clean, which are honest-sorry (deep or missing-Mathlib), and which definitions have been de-stubbed from earlier True/skeleton placeholders into genuine objects.*

D1. ChiralBundlePST.lean --- Chiral PST on Graph Bundles

The headline theorem of this file is a chiral generalization of the Bachman–Tamon bundle PST machinery (arXiv:1108.0339) that simultaneously contains, as the singleton-base case, the uniform-mixing speedup of Levine–Mesapam–Mustico–Tamon–Tucker–Zhan on chirally-signed complete graphs (arXiv:2605.04414). Concretely: a `ChiralBundle Q V` is a `GraphBundle` together with a chiral signing on each template edge and an intra-fiber signing on each fiber, satisfying Hermitian compatibility on reversed edges. The result `pst_iff_quotient_signed_pst` asserts that, when the bundle has regular fibers and biregular couplings, cell-uniform PST on the chirally-signed **total** graph is equivalent to PST on the chirally-signed **quotient** graph. The proof goes through the characteristic-isometry intertwining $S^* \exp(-i\tau A_{\text{signed}}) S = \exp(-i\tau A_{\text{quot, signed}})$ (Bachman–Tamon §2, extended chirally as in Levine et al. Lemma 2.1) plus the existing `EquitablePartition.pst_lift`.

The file’s substantive corollary is **contrapositive lifting**: ANY base graph Q whose vertex fibers carry the same K_n^σ signing inherits the Levine et al. $\frac{\pi}{3\sqrt{3}}$ uniform-mixing speedup at the cell-uniform level. Three concrete bundle families are spelled out: the chiral Hamming-attached path (`chiralHammingBundle n m`), the chiral Heawood template bundle (`chiralHeawoodBundle g n`), and a chiral bipartite bundle. A “phase-equitable refinement” lemma states that any chiral signing induces a (potentially finer) equitable partition by grouping vertices with the same outgoing-phase profile — the chiral analogue of Bachman–Tamon’s “twin partition.” Three open questions close the file: chiral product PST, sharpness of the $\frac{\pi}{3\sqrt{3}}$ time on K_n^σ , and chiral PGST on Heawood bundles.

Status. Carrier structures (`ChiralBundle`, the three concrete bundle families, the phase-equitable refinement) are concrete and typecheck; the headline intertwining `pst_iff_quotient_signed_pst` remains an honest sorry, reduced to the chiral characteristic-isometry lemma plus the proven spine `EquitablePartition.pst_lift`.

D2. BundlePSTLift.lean --- Universal Bundle PST Lift

This file gives the uniform iff statement `pst_iff_quotient` for cell-uniform PST on the total graph of a `GraphBundle` versus PST on its fiber quotient, and shows that it **strictly subsumes** the three classical product PST preservation theorems of Ge–Greenberg–Perez–Tamon (GGPT, arXiv:1009.1340). The Cartesian product, lexicographic product (regular second factor), and weak / tensor product all fall out as immediate corollaries. Beyond GGPT, the file states bundle-PST forms of products **not** covered by the original GGPT paper: the strong product $G \oplus H$, the conormal product, the disjunctive product, the `TemplateJoin` construction with constant fiber size, and `colorCompletion` of any coloring $V \rightarrow I$ over a complete template.

Two additional headline statements appear. The **stratified bundle lift** (`stratified_pst_lift`) extends the iff to bundles whose fibers fail regularity, by passing through a finer equitable partition adapted to extra symmetry. The **Bachman–Tamon–Feder reduction** (`cartesianProduct_quotient_naturality`) states that the Cartesian product of quotients is itself a quotient of the Cartesian product: a categorical naturality square between total and quotient. This

is, in particular, the Tower-5 ingredient that recurs in D9 below as a filtered-colimit preservation statement.

Status. The product corollaries land on the proven spine product bridge (Graphplay/Product/PST.lean, the GGPT Cartesian/lex/tensor cases); `pst_iff_quotient` is the honest-sorry universal statement, with the non-regular `stratified_pst_lift` and the naturality square as the remaining deep steps.

D3. ChiralGraphon.lean --- Chiral Graphons and Tower-4 Chiral Mixing

The Tower-4 graphon framework already permits complex-valued Hermitian kernels, so chirality is implicit; what was missing in the codebase, before this file, was the **interaction** of chirality with the graphon equitable partition. The file defines: (i) the chirality predicate `IsChiral W := mu times.circle mu (\{(x, y) : "Im"(W(x,y)) eq.not 0 \}) > 0`, making the real/imaginary decomposition explicit; (ii) measurable chiral signings `GraphonSigning` Ω as unimodular Hermitian-compatible measurable kernels, with entrywise action `Graphon.signedBy : Graphon -> GraphonSigning -> Graphon`; (iii) the graphon analogue of cell-cross-constant signings; (iv) the **chiral quotient** with explicit phase formula for cell pairs; (v) a gauge-theoretic interpretation in which cross-constant signings are flat $U(1)$ connections on the cell-partition graph (this is the seed of I7 below).

The headline statement is `chiralGraphonMixing_iff_quotientChiralMixing`: optimal cell-uniform chiral mixing on a graphon equitable partition is determined by a chiral phasing of the finite quotient. This is the Tower-4 quasi-infinite limit of Levine et al. Theorems 1–2. Two explicit limit families are recorded: the constant-phase graphon $K_n^\sigma \rightarrow \text{constantChiral}$ as $n \rightarrow \infty$, and the iterated-Hamming chiral $H(n, 4) \rightarrow \text{iteratedHammingChiral}$. Three open theorems close the file, including a conjectural Anantharaman–Sabri-style chiral connection in Benjamini–Schramm limits.

Status. The chirality predicate, measurable signings, and the chiral-quotient phase formula are de-stubbed into genuine measurable constructions; the analytic headlines (`chiralGraphonMixing_iff_quotientChiralMixing` and the two limit families) remain honest sorries, the most analysis-heavy of the dowsing-rod files, awaiting the Tower-4 measurable spectral-timing infrastructure.

D4. CoherentAlgebra.lean --- Tower 3 Commutative Equivalence

The guiding folklore equivalence here — apparently never written down formally — is: **an equitable partition of a graph G on V is the same data as a commutative unital $*$ -subalgebra of $\text{Matrix } V \mathbb{C}$ that contains the adjacency matrix $G.\text{adj}$ and is closed under the Schur (entrywise) product.** The file formalizes both directions of this dictionary. A `IsCoherent S` structure asserts that the submodule $S \subseteq \text{Matrix } V \mathbb{C}$ is unital, $*$ -closed, and closed under both matrix multiplication and the Schur product. The headline statement `equitablePartition_iff_coherentSubalgebraContaining` makes the equivalence precise: equitable partitions of G correspond bijectively to commutative coherent subalgebras containing $G.\text{adj}$.

A second contribution is the **Weisfeiler–Leman refinement chain** recast as a chain `WLAAlgebra G n : ℕ -> Submodule ℂ ("Matrix" V V CC)`, with the monotonicity `WLAAlgebra_mono`, the stabilization `WLAAlgebra_stabilises`, and the comparison `stableWL_le_orbitAlgebra` against the orbit algebra of $\text{Aut}(G)$. Theorem `BMAAlgebra_characterization` packages the Bose–Mesner algebra of an association scheme as the maximally symmetric case of a commutative coherent algebra. The file ends with a quantum-chromatic-number bridge `quantumChromatic_le_chromatic` that is the entry point to D5.

Status — largely proven. This is one of the most complete dowsing-rod files. The Schur-product algebraic structure (commutativity, associativity, J_n -unitality, bilinear-

ity, Hermitian compatibility) is de-stubbed and proven directly; the headline dictionary `equitablePartition_iff_coherentSubalgebraContaining` and the WL refinement chain (`WlAlgebra_mono`, `monotonicity / stabilization`) are proven. A single deep step remains an honest sorry (the maximal-symmetry Bose–Mesner identification). The earlier `*-algebra-closure` placeholders are now genuine `IsCoherent` obligations.

D5. NonCommutativeCoherent.lean --- Tower 3 Non-Commutative Quantum Equitable Partitions

D5 generalizes D4 to the operator-system / quantum-graph case (Duan–Severini–Winter arXiv:1002.2514, Mancinska–Roberson arXiv:1903.11491). A `QuantumGraph` is a unital `*-closed` subspace $S \subseteq M_{n(\mathbb{C})}$. A `QuantumEquitablePartition` $n \ S \ I$ is a unital `*-subalgebra` $A \subseteq M_{n(\mathbb{C})}$ containing S , together with a system of cell projectors $\{p_i\}_{i \in I}$ summing to 1, all lying in A — the non-commutative analogue of the partition projector Π_P . The **non-commutative coherent algebra** is then the smallest unital `*-subalgebra` containing S , equivalently the fixed point of non-commutative WL refinement (statement `WlFix_isCoherentAlgebra`).

The headline statement `QuantumEquitablePartition.pst_lift` says that PST on the quantum quotient implies cell-uniform PST on S . The quantum-chromatic-number characterization `quantumChromatic_le_iff_quantumHom` (Mancinska–Roberson) appears, as does the quantum Cayley graph generic-non-commutativity statement `quantumCayley_nonCommutative_generic`. Concrete examples: non-commutative K_n (off-diagonal operator system), quantum Hamming graphs as tensor powers, quantum Cayley graphs for non-abelian groups. The file closes with three open directions, including the GNW chain $\chi_f \leq \theta \leq \chi_q \leq \chi$ and quantum graphon limits.

Status — de-stubbed, headline lift proven. `QuantumHom` was hollow in the earlier draft; it now carries genuine `map_mul` and `map_trace` fields, and `QuantumHom.lifts_to_quotient` is proven from them (the quantum-graph-homomorphism lift). The non-commutative coherent algebra `ncCoherentAlgebra` is a genuine `sInf` of coherent algebras, proven closed (an `IsCoherentAlgebra`), and `WlFix` is genuinely coherent rather than a `True` placeholder. The remaining honest sorries are the quantum `pst_lift` spectral step and the generic-non-commutativity Cayley statement.

D6. FractionalRevivalNC.lean --- Fractional Revival on Coherent Algebras and Graphons

Chan–Coutinho–Tamon–Vinet–Zhan (arXiv:1907.04729) characterize fractional revival on graphs whose adjacency matrix lies in the Bose–Mesner algebra of an association scheme: this is the commutative Tower-3 case. This file extends FR theory to three increasingly non-classical settings. Tower 2: FR on a `WeightedGraph` via `IsFR` $G \cup v \ \tau \ \alpha \ \beta$ (matrix elements of $U(\tau)$ equal α at the diagonal u and β at the target v , everything else zero), with cell-uniform FR through equitable partitions and the lifting theorem `EquitablePartition.fr_lift`. Tower 3 non-commutative: FR between “cell-uniform state” projectors of a quantum graph (`Graphplay.QuantumGraph`), with spectral characterization `bose_mesner_fr_iff` and the commutative-reduction theorem `ncfr_commutative_reduction`. Tower 4 graphon: FR between bump-state classes on a measure space and the limit theorem `Graphon.fr_limit` bridging finite FR sequences to their graphon limits.

Three explicit families bring the development down to earth: FR on Cartesian products of cycles (Tamon clique example), FR on Hamming $H(n, q)$ via `hammingGraph_fr_iff`, and the **chiral K_n^σ fractional revival conjecture** `chiralKn_fr_conjecture` — the statement that the Levine et al. signing of K_n admits (α, β) -FR between any two distinct vertices for explicit $\alpha(n), \beta(n)$ that we conjecture extend the $\frac{\pi}{3\sqrt{3}}$ uniform-mixing time.

Status. The FR predicates and the three carrier settings (Tower 2/3/4) are concrete; the lifting theorems (EquitablePartition.fr_lift, ncfr_commutative_reduction, Graphon.fr_limit) and the chiral K_n^σ FR statement remain honest sorries at the spectral-characterization step.

D7. HypergraphPST.lean --- Three CTQW Models on Hypergraphs and the Doubly-Equitable Conjecture

CTQW on hypergraphs is unsettled: unlike on graphs, no single canonical Hamiltonian dominates. The file packages all three of the standard candidates as `WeightedGraph V` constructions: the **clique-expansion Laplacian** (replace each k -hyperedge with the complete graph K_k on its support); the **Hodge / incidence Laplacian** $L = BB^*$ formed from the complex vertex-edge incidence matrix of `Graphplay/Relational.lean`; and the **tensor / higher-order line walk** on the higher-order line graph whose vertices are k -tuples in a common hyperedge. The corresponding PST predicates `IsHypergraphPST_clique`, `IsHypergraphPST_hodge`, `IsHypergraphPST_tensor` are lifted through `RelEquitablePartition` quotients in parallel with `Graphplay/PST.lean`.

The headline conjecture is the **cross-model coincidence theorem** `cross_model_coincidence`: a hypergraph admits PST simultaneously in all three models if and only if its incidence relation satisfies a **doubly equitable** cell-uniformity condition (a `IsDoublyEquitable` structure that asks for equitability on both vertices and hyperedges). Concrete testable families: Steiner triple systems (`steinerTripleHodgePST_conjecture`), complete k -uniform hypergraphs (`completeKUniformPST_conjecture`), and partition designs from finite geometries. The file also lifts the chiral machinery: signed incidence, signed Hodge Laplacian, and `chiral_pst_lift_hodge`. A `Hypergraphon` (Tower-4 hypergraph) and its equitable partition close the file with `hypergraphon_pst_lift`.

Status. The three CTQW Hamiltonian constructions and the `RelEquitablePartition` quotient machinery are concrete and partly proven (several mixing/lift reductions discharge onto the spine); the cross-model coincidence conjecture and the design-family conjectures remain honest sorries.

D8. NoiseEquitable.lean --- Commutant Criterion and Caruso Noise-Assisted Speedup

A8 (`Graphplay/Toolkit/Noise.lean`) introduces the `NoiseModel` and the cell-uniform-symmetric preservation predicate at the statement level. D8 sharpens this. Q1 (preservation): we give a clean commutant-based criterion — every Lindblad jump operator L_k must lie in the **commutant of the partition algebra** — and characterize the **universal** case (preserves every equitable partition of every host) via `isUniversallyEquitable_iff_central`: the noise generators must lie in the center $\mathbb{Z}(M_{n(\mathbb{C})}) = \mathbb{C} \cdot 1$. Equivalently the only universally equitable noise is depolarizing. Q2 (Caruso noise-assisted speedup): we define a **breaking score** for a noise model as the L^2 distance of its commutant from the projector algebra of P , and state the **Caruso conjecture** `caruso_leading_order_speedup`: optimal noise-assisted speedup occurs at a non-trivial breaking score, strictly between fully symmetric and fully disruptive noise.

Three concrete models are analyzed for their breaking scores: depolarizing (universally equitable, score zero), site dephasing (equitable iff cells are vertex orbits, theorem `dephasing_cellUniformSymmetric_iff_orbit`), and boundary dephasing (not equitable for marked-refined partitions, explicit positive breaking score in `boundaryDephasing_breakingScore`). The file closes with `chiral_PST_open_mirror` and `openSystem_bachmanTamon`, the open-system analogues of the Bachman–Tamon closed PST quotient theorem, plus `ghost_symmetry_open_analogue`, an open-system version of the Bick–Sclosa ghost-of-symmetry phenomenon (arXiv:2110.13686).

Status. The commutant criterion and the three concrete noise-model breaking scores are concrete; the central-noise characterization is `UniversallyEquitable_iff_central` and the dephasing/orbit theorem are the proven anchors, while the Caruso speedup conjecture and the open-system Bachman–Tamon analogues remain honest sorries (deep / open-system spectral).

D9. FilteredColimitPST.lean --- Generalized Xie–Tamon: PST and Search Over Filtered Colimits

Xie–Tamon (arXiv:2301.07251, **No Infinite Tail Beats Optimal Spatial Search**) prove that the continuous-time spatial-search algorithm on K_n remains optimal under attachment of an infinite path P_∞ . The **categorical** reading of this result: the family $(K_n + P_m)_{m \geq 0}$, equipped with the distance-from- K_n equitable partition, is a **filtered diagram** in the partitioned weighted-graph category `WGraphP` (Tower 5, `Graphplay/Categorical.lean`). The `Quotient` functor preserves filtered colimits, and the PST predicate is continuous in operator norm; composing: PST holds in the limit quotient iff it holds with consistent times on every finite quotient.

The file states the general `ConsistentPartitionSequence.pst_inherited` theorem and exhibits a **concrete family table** with seven entries: $K_n + \text{path}$ (Xie–Tamon themselves); $K_n + \text{tree}$ (Bernard–Tamon–Vinet–Xie tail-tree, arXiv:2211.14704); $K_n + \mathbb{Z}^d$ multi-dimensional lattice; $K_n + \text{level-growth}$ geometric clique-shell expansion; `Hamming( $n, q$ )`-plus-path; `surface-Heawood-plus-path` (`surfaceHeawood_plus_path_quotient_stabilizes`); and $K_n \times \text{path}$ Cartesian. Three sharpenings: a **cofiltered** dual for infinite-state quantum Markov chains (`InversePartitionSequence.pst_lifted`); a **quantitative convergence-rate** refinement `ConsistentPartitionSequence.pst_rate_inheritance`; and three **failure-mode** conjectures (`failure_mode_unbounded_spectrum`, `failure_mode_continuous_spectrum`, `failure_mode_incoherent_times`) identifying obstructions to the limit lift. A **chiral** filtered colimit statement `ChiralConsistentPartitionSequence.pst_inherited` combines D1 with the present setup.

Status — master theorem proven. The headline `ConsistentPartitionSequence.pst_inherited` is proven, and the Xie–Tamon $K_n + \text{path}$ family is a genuine proven instance of it. The cofiltered dual `InversePartitionSequence` was formerly a hollow $P \rightarrow P$ identity; it is now a genuine inverse-limit structure with a real projective-stage quotient (`InversePartitionSequence.quotient`), though its lift theorem `pst_lifted` and the quantitative-rate refinement are honestly restated as sorries awaiting the convergence-rate spine lemma. The three failure-mode conjectures remain honest sorries by design.

D10. Conjecture93.lean --- The Flagship Falsifiable Conjecture

D10 is the flagship dowsing-rod result and we treat it in its own section below (see §3). In one sentence: a consistent sequence of finite weighted graphs (G_n, P_n) admits a graphon quasi-infinite limit and a cross-constant chiral signing speedup on the same partition if and only if the Bose–Mesner algebra of the family eventually coincides with the partition-projector algebra. This is **the first conjecture in the program that ties together Tower 2 (chiral), Tower 3 (Bose–Mesner / association schemes), and Tower 4 (graphons) in one falsifiable iff**.

Status — six genuine families, two halves honestly open. The six test families are now *genuine, distinct graph objects* (no sorry in any family def), each with an honestly-proven equitable partition: the chirally-signed complete graph K_n^σ , the genuine Hamming graph $H(n, q)$ on $\text{Fin } n \rightarrow \mathbb{Z}_q$ with single-coordinate-flip adjacency, the genuine Xie–Tamon $K_n + \text{path}$, the four-part complete multipartite $K_{n,n,n,n}$, the Heawood envelope, and the *real transposition Cayley graph* $\text{Cay}(S_n, \text{transpositions})$ — the last carrying a *proven* algebra-strictly-larger lemma (`cayleyS_n_algebra_strictly_larger`). The two halves of the conjecture itself,

conjecture93_weak_forward and conjecture93_weak_reverse, remain *genuinely open* (honest sorry); see §4.

2. Cross-Disciplinary Integrations (I1--I8)

*Each file under Graphplay/Integrations/ is a research-program seed pointing to a neighboring field where the spine applies and where the equitable-partition idiom is, we contend, currently underexploited. As with the dowsing-rod files, each item carries a **Status** line; several integration headlines have advanced from skeleton to de-stubbed or proven.*

I1. TQFT.lean --- Anyons, Modular Tensor Categories, and Surface Envelopes

The guiding observation: the Heawood envelope $K_{h(g)}$ of a graph embedded on a closed surface of genus g is a **discrete shadow** of the underlying surface. An anyonic system on that surface has Hilbert space organized as a representation of the modular tensor category (MTC) of the surface, and that representation decomposes into **topological sectors**. The file formalizes a ModularData A structure (Kitaev 2006, Appendix E) carrying the S -matrix, T -matrix, twists, and quantum dimensions, with the Verlinde formula verlinde and modular relations as sorried theorems. The headline result anyonic_equitable_partition says: when a surface graph carries an anyonic decoration, the corresponding sector decomposition **is** an equitable partition in the sense of Graphplay/Equitable.lean. Thus the Tower-3 quotient spine *is* the categorical sector decomposition, viewed combinatorially.

Three downstream results follow. surface_pst_isotopy_invariant makes PST on a surface graph isotopy-invariant (a topological criterion on the quotient). levinWen_groundstate_eq_cellUniform identifies the ground-state subspace of a Levin–Wen string-net Hamiltonian on a trivalent graph with the cell-uniform subspace of an explicit equitable partition. braid_gate_realizable shows that braid gates in topological quantum computation lift through the cell-uniform subspace, opening a path to **braid-quotient compilation** of topological-quantum-computation primitives. The Heawood–anyon match openQ1_HeawoodMatchesAnyons is left open: for each genus g the Heawood envelope predicts $h(g)$ vertices and the Reshetikhin–Turaev TQFT predicts a finite anyon count.

Status. ModularData, the Verlinde formula, and the sector-decomposition / equitable-partition correspondence are concrete; the headline anyonic_equitable_partition and the three downstream results remain honest sorries, and the Heawood–anyon match is open by design.

I2. RMT.lean --- Wigner Plus Equitable: Thermalizing-Yet-PST Hosts

The integration unexplored in the literature: classical random matrix theory studies the random bulk spectrum of $n \times n$ Hermitian matrices and identifies universal limits (Wigner’s semicircle, free convolutions); Graphplay’s Graphon.op_restrict_eq_quotient says that, **restricted to the cell-uniform subspace**, the spectrum of a graphon operator is exactly the deterministic spectrum of the finite quotient. Putting these together, a RandomGraphon R almost surely admitting a fixed equitable partition (captured by the AlmostSurelyEquitable structure) decomposes its spectrum as

$$\sigma(R.op) = \underbrace{\sigma(P.quotient)}_{\text{deterministic}} \sqcup \underbrace{\sigma(R.op|_{\text{cellUniform}^\perp})}_{\text{random Wigner bulk}}.$$

The headline thermalizing_yet_PST_host then declares: there exist random graphons whose orthogonal-to-quotient spectrum is GUE-distributed (hence thermalizing in the QECEC / ETH sense) while the cell-uniform sector remains deterministically PST-active — an explicitly engineered hybrid quantum host.

The technical scaffolding includes wignerGraphon_spectrum_semicircle (graphon analogue of Wigner’s law), wignerLimit_free_of_equitable (free convolution of the random part with the deter-

ministic quotient), and `graphon0p_is_free_semicircular` (the cell-uniform subspace is a free sub- W^* -probability space of the ambient W^* -probability space). Three open directions: WL-RMT universality (`open_problem_WL_RMT_universality`), a free-cumulant expansion of the quotient spectrum, and the quaternionic / GSE Tower-2 extension (`open_problem_quaternionic_graphplay`), the largest stated obstruction in the program.

Status. The spectral-decomposition scaffolding (`AlmostSurelyEquitable`, `wignerGraphon_spectrum_semicircle`, `graphon0p_is_free_semicircular`) is concrete; the headline thermalizing_yet_PST_host is now *honestly restated* — it names the genuine thermalizing host W_{bulk} with a real nonzero Wigner bulk and a deterministically PST-active cell-uniform sector, with the proof reduced to a single honest sorry at the GUE-spectrum step, rather than asserted vacuously.

I3. TensorNetworks.lean --- MERA Exact iff Iterated Equitable; Holographic Codes

A coarse-graining layer in a Multi-scale Entanglement Renormalization Ansatz (MERA) of Vidal is **literally an equitable-partition cell map**. The headline result `mera_exact_iff_equitable` makes this precise: a MERA is **exact** for a local Hamiltonian H iff each of its coarse-graining maps $c_n : V_n \rightarrow V_{n+1}$ is the cell map of an equitable partition P_n of the effective Hamiltonian H_n at level n . The disentangling unitaries handle the part of H_n that is **not** equitable; when there is no such part, they reduce to identities and the MERA captures the exact ground state. In categorical terms (matching `Graphplay.Categorical`), a MERA tower is a sequence of objects of `WGraphP` with morphisms going **down** under `Quotient`, i.e. a cochain whose bonding maps are the `Quotient.map` images of partition-respecting morphisms.

Downstream: `holographic_code_equitable` identifies holographic (Pastawski–Yoshida–Harlow–Preskill) quantum error-correcting codes as equitable bundles over bulk-boundary skeletons. `equitable_infinite_mera_has_limit` states the Tower-4 quasi-infinite limit for MERA towers — an explicit filtered colimit instance. `peps_exact_iff_two_d_equitable` extends to PEPS and `peps_on_surface` to surface PEPS. The compiler-facing `tensor_network_compiler` statement asserts that the MERA-of-an-equitable-host correspondence is **constructive**: given an equitable partition and its finite quotient, an exact MERA can be synthesized. Three open conjectures include `open_conjecture_gapped_equitable_tower` (gapped local Hamiltonians have MERA-exact equitable towers) and `open_conjecture_graphon_mera_limit` (the graphon limit of a MERA tower is the graphon-MERA of `Graphon/Limit.lean`).

Status — headline genuine. `IsExactMERA` was formerly a vacuous placeholder; it now carries an *independent* per-level equitability definition, so `mera_exact_iff_equitable` (and its holographic specialization `holographic_code_equitable`, plus the constructive `tensor_network_compiler` reverse direction) carry genuine content rather than being true by definition. The graphon-limit and gapped-tower conjectures remain honest sorries.

I4. OptimalTransport.lean --- Sinkhorn Respects Equitable; OT as a Design Primitive

A nonnegative real symmetric graphon is **almost** a transport plan: the only gap is marginal-correction, which Sinkhorn–Knopp / Cuturi address by row/column rescaling or entropic regularization. The integration bridge: equitable partitions of a graphon are **coarsened transport plans** (a finite quotient matrix $B : I \times I \rightarrow \mathbb{R}$ plus cell-uniform internal structure). The headline results `sinkhorn_preserves_equitable` and `sinkhorn_quotient_commutates` assert that Sinkhorn iterates preserve the cell-uniform subspace when the source is an equitable-partition graphon, and that the quotient gives a lower bound on the rate (`sinkhorn_rate_quotient_bound`). The Wasserstein-dis-

tance-via-quotient formula `wassersteinDistance_eq_quotient` says that, on cell-uniform support, the graphon-level W_2 reduces to a finite Wasserstein on the quotient measure.

The toolkit-facing claim is the **mixing-time** $\leftrightarrow$ **transport-rate** dictionary `mixing_sinkhorn_conjecture` and the constructive `quantum_sampler_existence`: given a target transport plan with an equitable structure, there exists a CTQW Hamiltonian whose cell-uniform evolution is the corresponding Sinkhorn iterate. This makes OT a **design primitive** for quantum sampler synthesis: the compiler emits, from a prescribed transport plan, a host CTQW whose uniform-mixing dynamics realizes the plan up to the $\log n/n$ entropic-mixing factor of the conjecture.

Status. The Sinkhorn/quotient carrier structure is concrete; the preservation and rate-bound headlines (`sinkhorn_preserves_equitable`, `sinkhorn_rate_quotient_bound`, `wassersteinDistance_eq_quotient`) and the constructive `quantum_sampler_existence` remain honest sorries at the entropic-mixing analysis.

I5. MeanFieldGames.lean --- Gao--Caines LQR as the Classical-Control Analog

Three threads share a mathematical primitive: graphon LQR control (Gao–Caines arXiv:2004.00677); mean-field control (Huang–Caines–Malhamé, Lasry–Lions); and graphon equitable partitions (Graphplay Tower 4). The underexploited observation: **the Gao–Caines invariant subspace is, in the most common applicable cases, exactly the cell-uniform subspace of an equitable graphon partition**. Their “Assumption (A5)” rests, in our vocabulary, on `GraphonEquitablePartition`. Once this identification is made, the Gao–Caines decomposition theorem reads as a **quotient LQR theorem**: the finite-dimensional quotient LQR on the cell index I is the entire content of the large-network LQR. The file states this as `equitable_LQR_reduction` with mild-solution existence and uniqueness (`mildSolution_exists_unique`) and `optimal_control_exists`.

The quantum-side translation: replacing LQR by Schrödinger dynamics (`schrodinger_cellUniform_invariant`) or Lindblad evolution (`lindblad_cellUniform_invariant`) keeps the same reduction. The `brachistochrone_reduction` statement asserts that the quantum brachistochrone problem on a host with equitable structure reduces to brachistochrone on the quotient — the entry point to a verified **pulse-synthesis** path for symmetric quantum chips. Three open directions: non-stationary equitable LQR, stochastic graphon equitable evolution, and chiral mean-field-game speedup (`chiral_MFG_speedup_open`).

Status — de-poisoned. The `MildSolution` predicate was formerly a vacuous `True` (under which *every* trajectory was a solution, poisoning the existence/optimality theorems); it now carries the genuine ODE-solution content (initial condition + `HasDerivAt` dynamics law). The cell-uniform invariance theorems (`schrodinger_cellUniform_invariant`, `lindblad_cellUniform_invariant`) quantify over *actual* solutions, and the Schrödinger law is the real `HasDerivAt` evolution. The remaining honest sorries are the LQR-reduction and brachistochrone optimality steps.

I6. Hodge.lean --- Equitable Hodge Decomposition and Harmonic Qubit Encoding

The combinatorial Hodge decomposition of Eckmann (1944/45) and the modern treatment of Lim (**Hodge Laplacians on graphs**, SIREV 2020) says that the cochain complex $C^0 \rightarrow C^1 \rightarrow \dots \rightarrow C^k$ of a simplicial complex decomposes any cochain into exact, coexact, and harmonic parts. The file sets up the `SimplicialComplex` and `Cochain` types, defines the `hodgeLaplacian` and the (sorried) `hodgeDecomp` and `harmonic_iso_cohomology` statements, and then introduces the file’s contribution: the `EquitableCochain` structure — an equitable partition of the cochain complex respecting the

coboundary — and the `hodgeQuotient` theorem asserting that the Hodge decomposition descends to the quotient simplicial complex of an equitable cochain.

Three downstream theorems wire this into the spine and beyond. `persistent_hodge_equitable` couples the equitable Hodge decomposition with persistent homology / TDA (Carlsson 2009): persistence diagrams of an equitable filtration are determined by the quotient filtration, opening **equitable speedup for TDA**. `harmonic_encoding_preserved` encodes a logical qubit in the harmonic subspace of an equitable cochain, with PST realized as a Hodge isomorphism (`hodgePST_lift`); this is a topological / homological avatar of cell-uniform PST. The harmonic-encoded qubit is intrinsically robust to noise restricted to the exact/coexact components, a complementary protection mechanism to the operator-algebraic noise analysis of D8.

Status — propagator real. The Hodge propagator `harmonicEvolve` is now the *genuine* operator exponential $\exp(-i\tau L_k)$ (`hodgePropFun`, the summed exponential series), not a schematic placeholder; `harmonicEvolve_eq_self` (harmonic cochains are fixed, derived from $L_k\varphi = 0$ via the proven series) is a real theorem rather than a definitional identity. The Hodge-Laplacian self-adjointness and coboundary-transpose lemmas are also proven; the `hodgeQuotient` descent and the TDA / encoding downstream results remain honest sorries.

17. `LatticeGauge.lean` --- Chiral = $U(1)$ Lattice Gauge; Cross-Constant = Flat; Hofstadter Quantization

The headline observations of this file are tight and verifiable. A `ChiralSigning V` is **exactly** a $U(1)$ lattice gauge field on the complete digraph on V : the equivalence `chiral_iff_ulGauge` is constructive (`U1GaugeField.toChiralSigning` and `ChiralSigning.toU1GaugeField`). The gauge field's **Wilson loop** gives its curvature, and the `ChiralSigning.CrossConstant` predicate of `Graphplay.Chiral` is equivalent to **cell-flatness** of the gauge field — vanishing curvature on every loop inside a single equitable cell. The cross-cell curvature depends only on the quotient cycle. This is the lattice gauge avatar of the chiral graphon gauge-theoretic interpretation of D3.

The two flagship consequences are the Hofstadter family results: `hofstadter_flux_quantization` says that when the Wilson loop around the bundle's monodromy cycle is the q -th root of unity $e^{i2\pi p/q}$, the cell-uniform spectrum of the bundle's adjacency lifts in **p/q quantized chunks**; this is the Hofstadter butterfly analog for graph bundles. The companion `hofstadter_chip_family` exhibits a family of equitable graph bundles whose flux quantization produces a constructive target spectrum. `equitable_hardware_design` then states the engineering corollary: **equitable hardware design corresponds to flat connection chip design**, realized in superconducting flux-qubit chips and synthetic-flux photonic lattices. The non-abelian extension to $U(N)$ matrix-valued signings is `matrix_gauge_field_preserves_equitable`, opening Yang–Mills-style lattice gauge integration with the spine.

Status — fully proven, zero sorry. This is the most complete integration file: the chiral $\equiv$ $U(1)$ -gauge equivalence (`chiral_iff_ulGauge`, with both transport maps), the cross-constant $\equiv$ cell-flatness characterization, the Hofstadter flux-quantization results, the engineering corollary `equitable_hardware_design`, and the non-abelian $U(N)$ extension all typecheck *without any sorry*. The chiral-as-gauge-field dictionary is therefore not a dowsing rod but a theorem.

18. `WLRefinement.lean` --- Equitable = WL Stable; PST = Phantom Symmetry

The Weisfeiler–Leman (WL) refinement chain $WL_1 \preceq WL_2 \preceq WL_3 \preceq \dots$ is the master algorithmic theory of graph isomorphism. This file threads it through `Graphplay`'s tower picture: 1-WL stable equals the orbit quotient by the coherent algebra (Tower 3 commutative dictionary, theorem `oneWL_stable_is_coarsest_equitable`); 2-WL generates the full coherent algebra

(coherentAlgebra_eq_2WL_span); k -WL gives higher-arity coherent configurations (Cai–Fürer–Immerman, with the strict-hierarchy theorem CFI_strict_hierarchy); WL on graphons converges in the cut metric to the graphon’s intrinsic equitable structure (graphonWL_limit_conjecture); the quantum WL chain QuantumWLStable reaches Mancinska–Roberson’s quantum-isomorphism graph algebra (MancinskaRoberson_qIsomorphism).

The two engineering corollaries are sharp. The **design budget** design_budget: the WL-stable partition is the **finest** equitable partition of G , so any equitable-partition-based PST scheme has at most $\#WL_{\infty(G)}$ cells — a hard upper bound on the quotient dimensionality of any PST protocol. The **phantom-symmetry** phantom_symmetry phenomenon pst_requires_WL_and_eigenSupport together with phantom_symmetries_exist asserts that PST is **more** than WL: there are non-isomorphic but WL-indistinguishable graphs that disagree on PST, witnessing that PST sees information beyond combinatorial colour refinement. Babai’s quasipolynomial GI bound (Babai_GI_quasipolynomial) gives the complexity-theoretic hook PST_design_sub_WL: equitable-PST design is a **sub-GI** problem.

Status. The WL-chain / coherent-algebra dictionary and the design-budget bound are concrete; the proven anchors are the structural WL-stability lemmas, while the graphon-WL cut-metric convergence, the Cai–Fürer–Immerman strict hierarchy, and the phantom-symmetry existence statement remain honest sorries (deep combinatorics).

3. Higher Towers (6--7)

Tower 6 --- Operator-Algebra-Valued Sheaves

Graphplay/Tower6.lean lifts the Tower-3 quantum-graph picture to a **sheaf of unital $\ast$ -algebras over a topological base space X** , together with a distinguished global adjacency section. Heuristically: a graph whose adjacency varies continuously as a parameter sweeps over X . The carrier objects are: a SheafGraph X (a sheaf of $\ast$ -algebras plus a global adjacency); a SheafEquitablePartition (an open cover trivializing the sheaf into cell algebras, together with a partition of the global adjacency respecting the cover); the **quotient sheaf** on the nerve of the cover; and the sheaf-level lift theorem sheaf_spectral_lift. The categorical global-section functor $\Gamma(X, \cdot)$ preserves filtered colimits (globalSection_preservesFilteredColimits), so Tower 5’s quasi-infinite limit lifts to the sheaf level.

Three lower towers are **recovered** as special cases. tower3_recovery: the constant sheaf on a one-point space yields a single quantum graph (Tower 3). tower4_recovery: the sheaf of $L^2(\Omega, \mu)$ -kernel-fibers over a measure space yields a graphon (Tower 4). tower5_recovery: the categorical projection lands in the partitioned weighted-graph category (Tower 5). The hardware-facing payoff is Hardware.driftRobust and robust_pst_neighbourhood: a PST protocol that holds on an open neighborhood of a parameter x_0 is robust under continuous hardware drift in x_0 , which is the operator-algebraic version of the noise-perturbation guarantee. adiabatic_is_section recasts the adiabatic theorem as a section of the sheaf graph.

Status — honest scaffold. Tower 6 is an honest scaffold of precise statements. The one fully-sorry-free anchor is the *skyscraper-sheaf* single-stalk constant-sheaf construction (a genuine TopCat .Sheaf UStarAlgCat X); the substantive sheaf_spectral_lift, the global-section colimit preservation, and the cell-uniform-stalk existence remain honest sorries, blocked on the missing Mathlib bundling of $\ast$ -algebra-valued sheaves (see §5). Earlier True-valued “hollow green” recovery statements have been narrowed to carry their genuine witnesses where available.

Tower 7 --- \$infinity\$-Categorical and Bicategorical Truncation

Graphplay/Tower7.lean is the highest abstraction tier in the program. The five-tower spine is recast in the language of stable ∞ -categories, in which the strict equalities of Tower 3 (idempotent partition projectors) are replaced by **coherent homotopies**. PST becomes an **equivalence in the unitary ∞ -groupoid** — PST “up to phase” upgrades to an honest isomorphism in the homotopy category, with phase data living as 2-morphisms. Mathlib does not yet carry a complete quasicategory library; the file therefore commits its content to the bicategory / $(2,1)$ -category level (Mathlib.CategoryTheory.Bicategory.Basic), where the statements actually typecheck. The InfinityCategory and StableInfinityCategory data are opaque placeholders; the bicategorical content is concrete.

The headline statements are: `infinity_pst_lift` (an infinity-categorical PST lift theorem); `bicategorical_lift` (the bicategorical truncation that **is** expressible in current Mathlib); `coherent_quasi_infinite_limit` (the filtered-colimit preservation theorem from Tower 5, lifted to the ∞ -coherent setting); `pst_as_Ext0` (PST as a derived Ext^0 computation in a derived Hermitian category); and `mtc_correspondence` (a recasting of modular tensor categories as $(\infty,1)$ -Tower-7 objects, connecting to I1 above). The flagship corollary is `topological_invariance_corollary`: under the topological-invariance conjecture `TopologicalInvarianceConjecture` (sorried), the ∞ -categorical PST predicate depends **only on the homotopy type of the base**, not on the underlying graph. The file is honest about its limit: **almost everything is sorried, because the supporting Mathlib library does not yet exist**.

Status — honest scaffold, unchanged frontier. Tower 7 remains the genuinely open higher tower. The `InfinityCategory` / `StableInfinityCategory` carriers are opaque placeholders and `infinity_pst_lift`, `pst_as_Ext0`, `coherent_quasi_infinite_limit`, and the topological-invariance corollary are honest sorries by design. The bicategorical truncation `bicategorical_lift` is the highest layer that typechecks today; the previous draft’s stray `True`-valued conveniences have been converted to honest sorries so the file no longer reads as “trivially proven” where it is in fact awaiting an ∞ -categorical Mathlib (see §5).

4. The Flagship Conjecture 9.3

Statement

Let $\mathcal{S} = (G_n, P_n)$ be a consistent sequence of finite weighted graphs with equitable partitions of fixed cell index type I and consistent embeddings $\iota_n : V_n \hookrightarrow V_{n+1}$ respecting cell labels (a `ConsistentPartitionSequence I` in Graphplay/Graphon/Limit.lean). Define:

- `admitsGraphonLimit( $\mathcal{S}$ )`: there exists a graphon W_∞ on (Ω, μ) with an equitable partition P_∞ of the same index type, such that the finite quotient matrices $\mathcal{S}.\text{quotient}(n)$ converge to $P_\infty.\text{quotient}$.
- `admitsChiralSpeedupOnPartition( $\mathcal{S}$ )`: there exists a sequence $\{s_n\}$ of nontrivial chiral signings, each `CrossConstant` on $\mathcal{S}.\text{cells}(n)$, consistent under the embeddings (phases pull back), whose signed quotient exhibits a strictly faster cell-uniform mixing / PST time than the unsigned quotient.
- `eventuallyAlgebraCoincidence( $\mathcal{S}$ )`: there exists N such that for every $n \geq N$, the smallest coherent algebra containing $(G_n).\text{adj}$ equals the partition-projector algebra of $\mathcal{S}.\text{cells}(n)$.

The flagship conjecture, in its **weak** (cross-constant) form, is the iff:

$$\text{admitsGraphonLimit}(\mathcal{S}) \wedge \text{admitsChiralSpeedupOnPartition}(\mathcal{S}) \iff \text{eventuallyAlgebraCoincidence}(\mathcal{S}).$$

The Lean statement is `Conjecture93_weak` in `Graphplay/Dowsing/Conjecture93.lean`, decomposed into the forward direction `conjecture93_weak_forward` (algebra coincidence implies both halves) and the reverse direction `conjecture93_weak_reverse` (both halves imply algebra coincidence). The **strong** form `Conjecture93_strong` replaces the cross-constant signing constraint by an arbitrary unitary signing; this version we conjecture is **false**, with the candidate counterexample described below.

Why It Is Interesting

This is the **first conjecture in the program that ties together Tower 2 (chiral), Tower 3 (Bose–Mesner / association schemes), and Tower 4 (graphons) in one falsifiable iff**. The forward direction says that a graphon-limit family that also admits a chiral speedup is **rigid enough** to behave like an association scheme: it pins down the precise sense in which the Levine et al. (arXiv:2605.04414) chiral speedup is more than an arbitrary unitary perturbation. The reverse direction is the association-scheme statement: a family that is uniformly an association scheme automatically admits both a graphon limit (the natural step-graphon of the scheme classes) and a chiral speedup (any cross-constant signing on the quotient).

The conjecture is consistent with — and strictly strengthens — the fractional-revival-iff-Bose-Mesner picture of Chan–Coutinho–Tamon–Vinet–Zhan (arXiv:1907.04729). If true, it gives a clean operational characterization of **which families are “scheme-like” in the limit**: exactly those for which the Bose–Mesner algebra is generated by partition projectors.

Worked Examples (Six Test Families)

`Graphplay/Dowsing/Conjecture93.lean` records six families covering the expected positive and negative cells of the truth table. *Each family is now a genuine, distinct graph object* — no family `def` contains a `sorry`, and each carries an honestly-proven equitable partition (regularity for the single-cell families, the genuine part partition for $K_{n,n,n,n}$, the conjugacy structure for Cayley). This is a correction over an earlier draft in which several families were complete-graph stand-ins. We summarize each:

4.1 K_n^σ with constant phase (both halves hold). Trivial single-cell partition; Bose–Mesner algebra = $\mathbb{C} \cdot I + \mathbb{C} \cdot J$ equals the partition-projector algebra. The Levine et al. K_4 unitary signing is cross-constant on the unique cell pair.

4.2 Hamming $H(n, q)$ with distance partition (both halves hold). Genuine graph on $\text{Fin } n \rightarrow \mathbb{Z}_q$ with single-coordinate-flip adjacency, proven regular. Algebra coincidence exact on every class. Any nontrivial phase on a single distance class gives a cross-constant speedup signing.

4.3 $K_n + \text{path}_n$ Xie–Tamon family (both halves false). Graphon limit exists by explicit Xie–Tamon construction. However, the Bose–Mesner algebra of $K_n + P_n$ strictly contains the partition-projector algebra: the tridiagonal path tail has algebra dimension $n + 1$, while the cell-pair-constant algebra on the distance partition has lower dimension. The conjecture predicts no chiral speedup on this partition, and indeed every cross-constant signing acts trivially or breaks the partition. **The cleanest both-sides-false witness, and the family that motivated the conjecture.**

4.4 $K_{n,n,n,n}$ four-color completion (both halves hold). Color partition; rank-2 Bose–Mesner algebra equals the projector algebra; arbitrary quotient phases descend.

4.5 Heawood envelope $g \rightarrow \infty$ (both halves hold trivially). $K_{H(g)}$ with single-cell partition reduces to K_n .

4.6 $\text{Cayley}(S_n)$ by transpositions (both halves false). Genuine transposition Cayley graph (the identity is *not* adjacent to a 3-cycle; the adjacency is honestly non-constant), proven regular by left translation. Conjugacy partition. The Cayley Bose–Mesner contains the full group algebra $\mathbb{C}[S_n]$ of dimension $n!$.

while the projector algebra has dimension $p(n)^2$. For $n \geq 5$ these differ — and *this strict-larger gap is itself proven* (cayleyS_n_algebra_strictly_larger), making this the one family whose negative cell is a theorem rather than a sorry. Any nontrivial cross-constant signing on conjugacy classes is either trivial or breaks the partition.

Best Guess About Truth

The weak conjecture is probably true. Three reasons: **(i)** No known explicit counterexample: every family checked falls into the (LHS true, RHS true) or (LHS false, RHS false) cells; the impossible (LHS true, RHS false) case is unfilled. **(ii)** Structural reason for the forward direction: if (G_n, P_n) admits a chiral speedup on the partition, then by `chiral_mixing_optimization` the speedup is governed by a chiral phasing of the quotient, living in the partition-projector algebra. Combined with graphon-limit commutativity (limit operator commutes with cell projection), this should force algebra coincidence in the limit. **(iii)** 1907.04729 precedent: fractional revival on graphs from association schemes (the canonical “both halves hold” case) is exactly where the conjecture is known to hold by the literature.

The strong conjecture is probably false. The cleanest candidate counterexample uses a **non-cross-constant** unitary signing on $K_n + \text{path}_n$ that introduces a phase-equitable refinement of the distance partition (cf. the `signedBy_phaseRefined_equitable` lemma of D1). Such a signing can produce a strictly faster cell-uniform mixing time without making the underlying algebra match the projector algebra. The explicit construction is open; the locus is identified.

Next Steps

To prove (weak): discharge `tower3_equitable_partition` (Lemma 2 of 1907.04729) to characterize `partitionProjectorAlgebra` as $\{M : \Pi_P M = M \Pi_P \wedge \Pi_P M \Pi_P = M\}$; prove `Bose-Mesner.isCoherentAlgebra` and the commutativity lemma; then run the forward direction through `chiral_mixing_optimization` intertwining and the reverse direction through graphon convergence.

To refute (weak): construct a partition sequence whose Bose–Mesner is strictly larger than the partition-projector algebra at every stage, but for which some cross-constant chiral signing still produces a strict cell-uniform mixing improvement. The likeliest hunting ground: **non-vertex-transitive graphs with non-symmetric distance distributions** — distance-regular-but-not-distance-transitive bipartite double covers of non-self-paired association schemes.

To refute (strong): take any both-sides-false family and search for a unitary signing that breaks `CrossConstant` but still descends through a phase-refined partition. The chiral half becomes true under the strong definition while RHS stays false — a strong-version counterexample.

The concrete first computation we recommend is the Bose–Mesner algebra dimension of $K_n + P_n$ vs its distance-projector algebra dimension, in Lean, for $n = 3, 4, 5$. If the dimensions differ as predicted, that nails down family 4.3 as a both-sides-false instance, providing the first nontrivial confirmation of the conjecture.

5. New Results Extending the Program

*This revision records four developments that extend the catalog past the original D/I/Tower list. The first two are **applications** (under `Graphplay/Applications/`), the third is a **verified compiler** (under `Graphplay/Integrations/`), and the fourth is a **new dowsing-rod file** (under `Graphplay/Dowsing/`) opening the relativistic frontier with its combinatorial band-touching half proven and its analytic continuum half an honest sorry.*

5.1 SparseSearch.lean --- Quadratic Search on a Buildable Host

The flagship quantum-advantage result of the program (Grover / CTQW $O(\sqrt{N})$ search) lived on the complete graph K_n , which needs $n(n-1)/2$ all-to-all couplings and is unbuildable at scale. Graphplay/Applications/SparseSearch.lean carries the same advantage to the Boolean hypercube Q_d ($N = 2^d$ vertices, degree only $d = \log_2 N$ — the standard sparse PST/search-chip family of Childs–Goldstone, quant-ph/0306054). The mechanism is a *pure equitable-partition spine fact*: with marked vertex w , the Hamming-distance-from- w partition into the $d+1$ cells is equitable for the search Hamiltonian $H = -\gamma A(Q_d) - |w\rangle\langle w|$, collapsing the dynamics onto a $(d+1)$ -dimensional weighted path with binomial couplings. The *deliverable is axiom-clean*: the binomial branching hamming_branching, the equitable partition hammingPartition_equitable, its $d+1$ cell count, the d -regularity / sparsity (hypercube_degree, hypercube_sparse), and the collapsed-chain quotient reduction search_quotient_reduction are all proven. The $O(\sqrt{N})$ running-time clause leans on the CNO spectral-ratio criterion and is an honest sorry (the single deep spectral-timing step). The file further records the d -dimensional periodic lattice $\mathbb{Z}_L^d$ as the buildable-but-usually-non-advantageous host, with the *dimensional threshold* $d > 4$ for optimal $\Theta(\sqrt{N})$ search (the $d = 4$ marginal case loses a $\sqrt{\log N}$ factor, tied to the convergence threshold of the lattice Green’s function $\sum_k 1/(1 - \cos k)$); the strongly-regular case is recorded as an honest sorry (strongly_regular_sparse_search).

5.2 CompileML.lean --- A Falsifiable Chip Experiment

Where Graphplay/Applications/IBMHeavyHex.lean runs the *disassembly* direction (chip $\rightarrow$ equitable quotient $\rightarrow$ CTQW primitive, with the axiom-clean heavy-hex 2×2 quotient and heavyHex_pst_lift), Graphplay/Applications/CompileML.lean runs the *inverse*, compilation direction: an ML primitive (a two-class structured-attention / associative-recall task, where recall = perfect transfer between a query-uniform and a key-uniform state) compiles to native IBM-Heron coupling weights plus a static CTQW schedule via compileToHeavyHex : MLTarget $\rightarrow$ "HardwareSpec" times "Schedule" (and a Majorana-tetron variant compileToMajorana). The compilation-correctness theorems (compileToHeavyHex_wellFormed, compiled_cellUniform_realizes_target, compiled_schedule_realizes_target) are genuinely proven, reusing the axiom-clean heavy-hex PST lift. The payoff is compiled_experiment_prediction: a *falsifiable lab prediction* — prepare the query-uniform state, run the emitted schedule for $t = \pi/(2q)$ ($q = 2\sqrt{N-1}$), and the key population follows the predicted amplitude-squared curve, with ideal recall at the compiled time under zero-residual noise. This is the program’s first end-to-end *compile $\rightarrow$ predict* statement handed to hardware.

5.3 TransformerDSL.lean --- The Functor Stack as a Verified Compiler

Graphplay/Integrations/TransformerDSL.lean realizes the program’s headline programming-language thesis *as a proven theorem, with zero sorry*. It builds a small finite typed DSL TransformerProgram V (five constructors: attention, multiHead, add, residual, feedForward, plus compose for layer stacking), gives it a denotation into the WeightedGraph quantum-walk semantics, gives it a *compiler* that lowers a block-equitable program to its small $r \times r$ quotient through the Quotient functor of Graphplay.Categorical, and *proves* the compiler semantics-preserving: the compiled (quotient) meaning is exactly the denotational meaning restricted to the cell-uniform subspace. The slogan — “WGraphP morphisms are typed programs; the Quotient functor is the optimizing compiler pass; PST / search / mixing are the runtime primitives” — is, for this DSL, no longer a metaphor but a verified ML $\rightarrow$ quantum compiler.

5.4 DiracLimit.lean --- Dirac / Relativistic Enrichment

The catalog’s natural next enrichment is *relativistic*: the discrete-time coined walk is, in the appropriate continuum limit, governed by a Dirac rather than a Schrödinger operator, and a bipartite-equitable host should limit to a *Dirac cone* (linear band crossing) rather than a

parabolic band. `Graphplay/Dowsing/DiracLimit.lean` formalizes this frontier as a new dowsing-rod file. It carries the Pauli/Clifford layer (proven Hermitian Pauli matrices, $\sigma_x^2 = 1$, the `CliffordWeightedGraph` block Hamiltonian and its proven Hermiticity), a `BlochHamiltonian` band-structure layer, and the off-diagonal Bloch model whose two-band eigenvalues $\pm |f(k)|$ are computed exactly (`offDiagonalBloch_eigenvalues`) with the band-touching criterion $f(k_0) = 0$ proven (`offDiagonalBloch_band_touch`, `offDiagonalGap_eq_zero_iff`).

Status — de-stubbed, band-touching proven, continuum limit honest-sorry. Contrary to an earlier draft that recorded this as “not yet present in Lean,” the file now exists and the *combinatorial / spectral half is proven axiom-clean*: `honeycombFormFactor_dirac_point` (the honeycomb form factor vanishes at the Dirac point), the headline `bipartite_equitable_dirac_cone` (a bipartite-equitable host’s off-diagonal quotient *has* a band-touching Dirac point), and its worked example `heavyHex_hosts_dirac_cone` (the IBM heavy-hex quotient hosts a Dirac cone) all discharge their band-touching content from the proven off-diagonal layer. What remains an honest sorry — flagged -- BLOCKED: inline — is the genuinely analytic *continuum half*: the linear-slope / small- k Taylor expansion certifying *linear* (not merely touching) dispersion, and the coined-walk $\rightarrow$ Dirac-operator scaling limit `coinedWalk_continuum_is_dirac`, for which Mathlib has no scaling-limit calculus. The Schrödinger-side coined-walk infrastructure is `Graphplay/StdLib/CoinedWalk.lean`; the relativistic enrichment now sits beside it. The remaining analytic step joins the open-boundary list of §6.

6. Open Boundary: What Is Not Yet Formalized

The program has identified **open boundary** obstructions where the spine predicts content but the supporting Mathlib (or, in some cases, mathematical) infrastructure does not yet support the proof. We list the three largest.

Quaternionic / GSE Tower 2. The Tower-2 weighted-graph layer uses complex Hermitian adjacencies. The quaternionic / symplectic analogue — a Tower-2 layer over $\mathbb{H}$ with self-conjugate adjacency — would extend chiral signings to the Gaussian Symplectic Ensemble (GSE) and connect to the I2 RMT integration via `gse_quaternionic_graphon_open`. The blocker is not conceptual but engineering: Mathlib does not currently carry quaternionic matrix infrastructure at the level of completeness of the complex case. The statement `open_problem_quaternionic_graphplay` flags the right type to define and the right theorems to port, but the proofs depend on a Mathlib quaternionic-linear-algebra library that does not exist.

∞ -categorical Mathlib blockage. Tower 7 lives at the bicategorical / $(2, 1)$ -truncation precisely because Lean+Mathlib does not yet support quasicategories or stable ∞ -categories. The ∞ -categorical PST lift `infinity_pst_lift` and the derived Ext^0 formulation `pst_as_Ext0` are recorded as Props about hypothetical structures with sorry-ed existence, awaiting (i) a Mathlib quasicategory library, (ii) a derived-category framework, and (iii) a $*$ -enriched stable ∞ -category to host the unitary ∞ -groupoid. The bicategorical truncation `bicategorical_lift` is the highest layer that typechecks today.

Sheaf-of- $$ -algebras infrastructure.* Tower 6 needs a category `UStarAlgCat` of unital $*$ -algebras over $\mathbb{C}$, ideally the Mathlib bundled `AlgebraCat  $\mathbb{C}$` carrying a `StarRing` instance naturally. The Mathlib `AlgebraCat  $\mathbb{C}$` bundle is not currently star-equipped, forcing the `Graphplay.Tower6` file to maintain its own bundle. The `SheafGraph` and `SheafEquitablePartition` machinery is fully written; the existence-of-cell-uniform-stalks proofs depend on the Mathlib bundling that is being deferred.

Lesser open boundaries identified throughout the program: graphops (Backhausz–Szegedy generalization of graphons by removing the L^2 symmetric-kernel assumption; the quotient spine survives when the operator commutes with the projection); open-system Lindbladian extensions of the Bachman–Tamon quotient theorem (`D8 openSystem_bachmanTamon`); the non-commutative coherent-algebra

GNW chain $\chi_f \leq \theta \leq \chi_q \leq \chi$ (D5 `OpenDirection.GNW_chain`); time-dependent / non-stationary equitable partitions (I5 `nonstationary_equitable_LQR_open`); and the analytic *continuum half* of the Dirac / relativistic enrichment of §5.4 — the coined-walk $\rightarrow$ Dirac-operator scaling limit (`coinedWalk_continuum_is_dirac`) and the linear-slope dispersion certificate — whose combinatorial band-touching half is now proven in `Graphplay/Dowsing/DiracLimit.lean` but whose scaling-limit calculus Mathlib does not yet support.

7. Reading Guide

For readers wanting to navigate the codebase, we map content to file locations.

Spine (already in main paper). `Graphplay/Weighted.lean`, `Graphplay/Equitable.lean`, `Graphplay/Spectral.lean`, `Graphplay/PST.lean`, `Graphplay/Mixing.lean`, `Graphplay/Search.lean`, `Graphplay/QuantumGraph.lean`, `Graphplay/Chiral.lean`, `Graphplay/Bundle.lean`, `Graphplay/Categorical.lean`, `Graphplay/Graphon.lean` and the `Graphplay/Graphon/*` files.

Dowsing-rod (this companion §1).

- `Graphplay/Dowsing/ChiralBundlePST.lean` — D1
- `Graphplay/Dowsing/BundlePSTLift.lean` — D2
- `Graphplay/Dowsing/ChiralGraphon.lean` — D3
- `Graphplay/Dowsing/CoherentAlgebra.lean` — D4
- `Graphplay/Dowsing/NonCommutativeCoherent.lean` — D5
- `Graphplay/Dowsing/FractionalRevivalNC.lean` — D6
- `Graphplay/Dowsing/HypergraphPST.lean` — D7
- `Graphplay/Dowsing/NoiseEquitable.lean` — D8
- `Graphplay/Dowsing/FilteredColimitPST.lean` — D9
- `Graphplay/Dowsing/Conjecture93.lean` — D10 and §4 above; see also `paper/conjecture93_notes.md` for the detailed worked-example notebook.

Integrations (this companion §2).

- `Graphplay/Integrations/TQFT.lean` — I1
- `Graphplay/Integrations/RMT.lean` — I2
- `Graphplay/Integrations/TensorNetworks.lean` — I3
- `Graphplay/Integrations/OptimalTransport.lean` — I4
- `Graphplay/Integrations/MeanFieldGames.lean` — I5
- `Graphplay/Integrations/Hodge.lean` — I6
- `Graphplay/Integrations/LatticeGauge.lean` — I7
- `Graphplay/Integrations/WLRefinement.lean` — I8

Higher towers (this companion §3).

- `Graphplay/Tower6.lean` — sheaves of $*$ -algebras over a topological base; recovers Towers 3, 4, 5 as point-base, measure-base, and global-sections cases.
- `Graphplay/Tower7.lean` — ∞ -categorical / bicategorical truncation; the topological-invariance corollary.

New results (this companion §5).

- `Graphplay/Applications/SparseSearch.lean` — §5.1, hypercube Q_d quadratic search and the $d > 4$ lattice threshold.
- `Graphplay/Applications/CompileML.lean` — §5.2, the falsifiable compile-to-chip experiment (with `Graphplay/Applications/IBMHeavyHex.lean` for the disassembly direction).
- `Graphplay/Integrations/TransformerDSL.lean` — §5.3, the verified (zero-sorry) functor-stack $\text{ML} \rightarrow \text{quantum compiler}$.

- Graphplay/Dowsing/DiracLimit.lean — §5.4, the Dirac / relativistic enrichment (proven band-touching bipartite_equitable_dirac_cone / heavyHex_hosts_dirac_cone; honest-sorry continuum scaling limit coinedWalk_continuum_is_dirac), beside the Schrödinger-side Graphplay/StdLib/CoinedWalk.lean.

Conjecture 9.3 apparatus. Graphplay/Dowsing/Conjecture93.lean carries the full statement-level infrastructure (Conjecture93_weak, Conjecture93_strong, the forward and reverse implications, the six test families, the sufficient conditions, the failure modes, the chiral / strong distinction). The discursive notes are in paper/conjecture93_notes.md. The status table is summarized in paper/quasi_infinite_adjoint_v3.typ §9.

Cross-references. Spine theorems consumed by this companion include EquitablePartition.pst_lift (Bachman–Tamon), Graphon.op_restrict_eq_quotient (Tower 4 spectral lift), Quotient.preservesFilteredColimits (Tower 5 universal Xie–Tamon), Chiral.signedBy_preserves_equitable (Tower 2 chiral lift), and coherentAlgebra and BoseMesner (Tower 3 algebras).

Everything below the spine is, intentionally, an unfinished research program. The companion’s purpose is to make precise both what would be gained and what would have to be proved.